

Numerical Analysis II

Xiaojing Ye, Math & Stat, Georgia State University

Spring 2019

About Instructor

- ▶ **Instructor:** Prof. Xiaojing Ye
- ▶ **Email:** xye@gsu.edu
- ▶ **Webpage:** <https://math.gsu.edu/xye>
- ▶ **Office:** 25 Park Place, Room 1436
- ▶ **Office hour:** MW 4:30–5:30PM (other time by appointment only)

About Course

- ▶ **Time:** MW 2:00–3:15PM
- ▶ **Location:** 427 Sparks Hall
- ▶ **Textbook:** Numerical Analysis (10th edition). Authors: Burden and Faires. Publisher: Brooks/Cole, Cengage Learning
- ▶ **Prerequisite:** Grade of C or higher in Math 2215 Multivariate Calculus or equivalent and the ability to program in a high-level language.

Course Topics

- ▶ **Numerical methods for ordinary differential equations:** Euler, modified Euler, Runge-Kutta methods, explicit and implicit methods, predictor-corrector method
- ▶ **Direct methods for solving linear systems:** Gaussian elimination and pivoting strategies, matrix factorizations.
- ▶ **Iterative methods for linear systems:** Matrix theory, Jacobi, Gauss-Seidel, SOR, conjugate gradient methods.
- ▶ **Approximation theory:** linear and polynomial least squares, Legendre and Chebyshev polynomials, rational function approximations, trigonometric polynomial approximations.

Homework

- ▶ 8 Homework assignments throughout the semester
- ▶ 2 or 3 points each. Total 20 points
- ▶ Homework due at the beginning of the lecture on due day
- ▶ Individual work.
- ▶ **Late homework will not be accepted**

Exams

- ▶ **Midterm 1 (25 pts):** in class on February 13.
- ▶ **Midterm 2 (25 pts):** in class on March 13.
- ▶ **Final (30 pts):** 13:30-15:30PM on May 1.

Grading Policy

Grading points:

Homework	20 pts
Midterm 1	25 pts
Midterm 2	25 pts
Final	30 pts
Total	100 pts

Grading scale:

Score	0–59	60–69	70–76	77–79	80–82	83–86	87–89	90–92	93–96	97–100
Grade	F	D	C	C+	B–	B	B+	A–	A	A+